SURV/SURVMETH 720

Sources of Measurement Error

Class #9

November 10, 2025

Last week: Methods of Studying Measurement Error

1. Question evaluation methods

2. Using measures external to the survey

3. Randomized assignment to different procedures

4. Repeated measures

Table 7.2 Percentage of Respondents Forbidding and Not Allowing Speeches in Favor of Communism

Do you think the United States should forbid speeches in favor of communism?		Do you think the United States should allow speeches in favor of communism?	
Yes (forbid)	39.3%	No (not allow)	56.3%
n	(409)	*n*	(432)

Source: H. Schuman and S. Presser, *Questions and Answers in Attitude Surveys*, Academic Press, Orlando, 1981, Table 11.2, p. 281.

- Are the two Q forms measuring the same concept but subject to different response errors? Or are two different concepts being measured?

- What does (56.3 - 39.3) represent?

Contrast this to…

-

What is the difference with the earlier example ?

Some common indicators of Measurement Bias

- Socially desirable responding
  - Assumption is that procedure that solicits more reports of socially undesirable behaviors or fewer reports of socially desirable behaviors is better
- Satisficing
  - Assumption: procedure that reduces satisficing behaviors is better (e.g., less missing data, fewer rounded answers, reduced straightlining on grids, etc.)

Implicit measurement error model

-

y_{ij} Survey response by respondent *i* using method *j* (e.g. question wording)

X_i True value of characteristic for the *i*th person

M_j Error associated with *j*th method (e.g., question delivery – note that the effect can vary by sub-groups)

ε_{ij} Deviation of the *i*th person from the average effect of the *j*th method

Potential Limitations of Randomized Designs

- Experimental groups that are not equivalent
  - e.g., IWER group 1 assigned to deliver wording 1 but are generally less successful in recruiting *(also see West and Olson, 2010).*
- Limited external validity
  - If, e.g., IWERs/survey ops aware of the experiment
  - measurement conditions cannot be replicated outside of experiment

Methods of Studying Measurement Error

1. Question evaluation methods

2. Using measures external to the survey

3. Randomized assignment to different procedures

4. Repeated measures

Reinterview Studies

Involves repeated measurements of same underlying construct at different time points (can also do within the same survey but obvious issues)

- Works for for factual and behavioral questions

What is this studying? Bias or variance?

Assumptions (always watch!) of Reinterviews to Measure Response Variance

- No change in true values over trials
  - threatened by long time between trials
- No memory effect of first trial on the response to the second trial
  - threatened by short time between trials
- Other essential survey conditions the same between trials
  - threatened by using supervisors as interviewers

Today's Agenda

Discussion of different sources of measurement error:

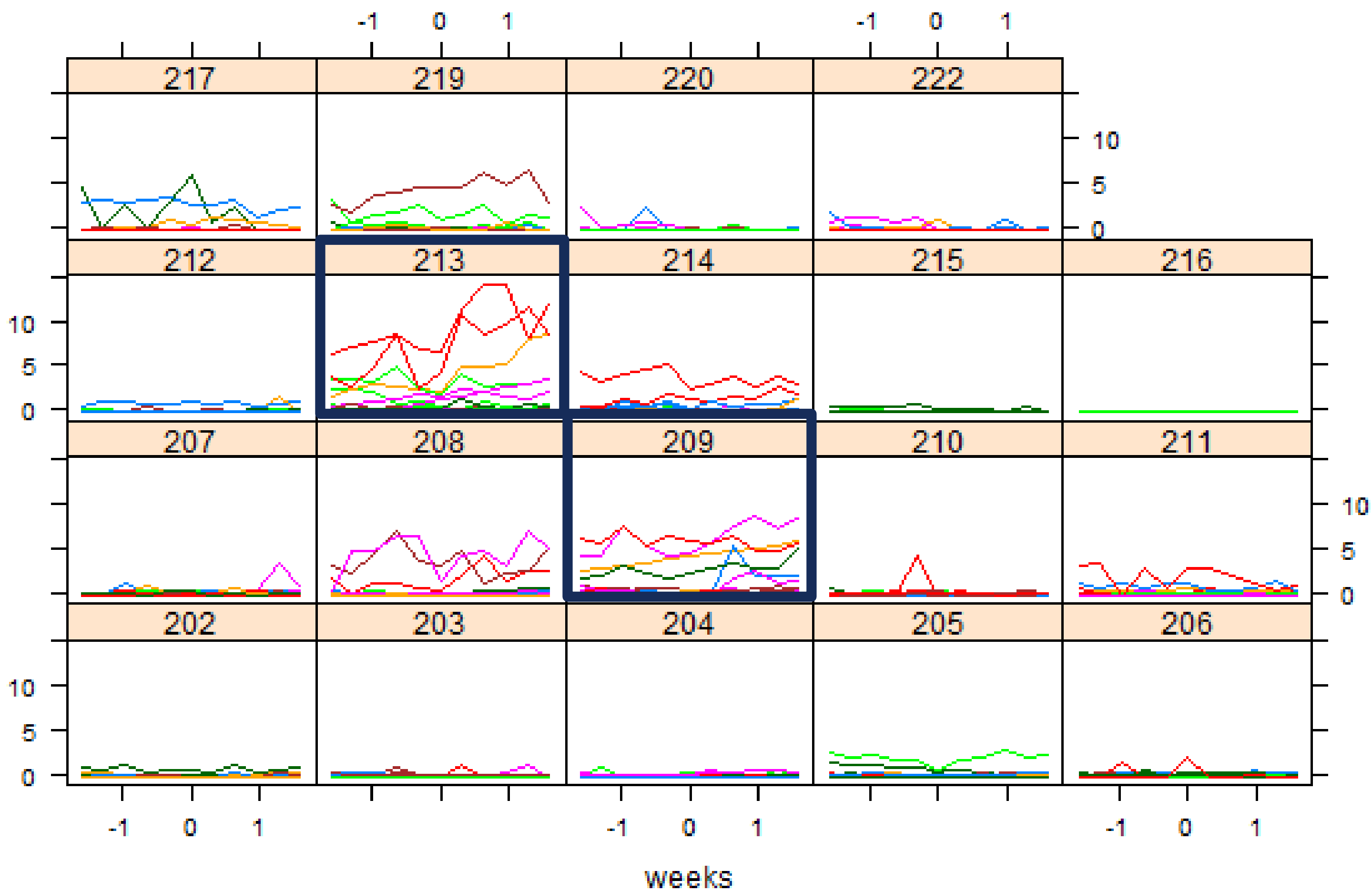
- Interviewers
- Respondents
- Instrument/Questionnaire
- Mode

Interviewers

Many forms of interviewer errors leading to measurement error

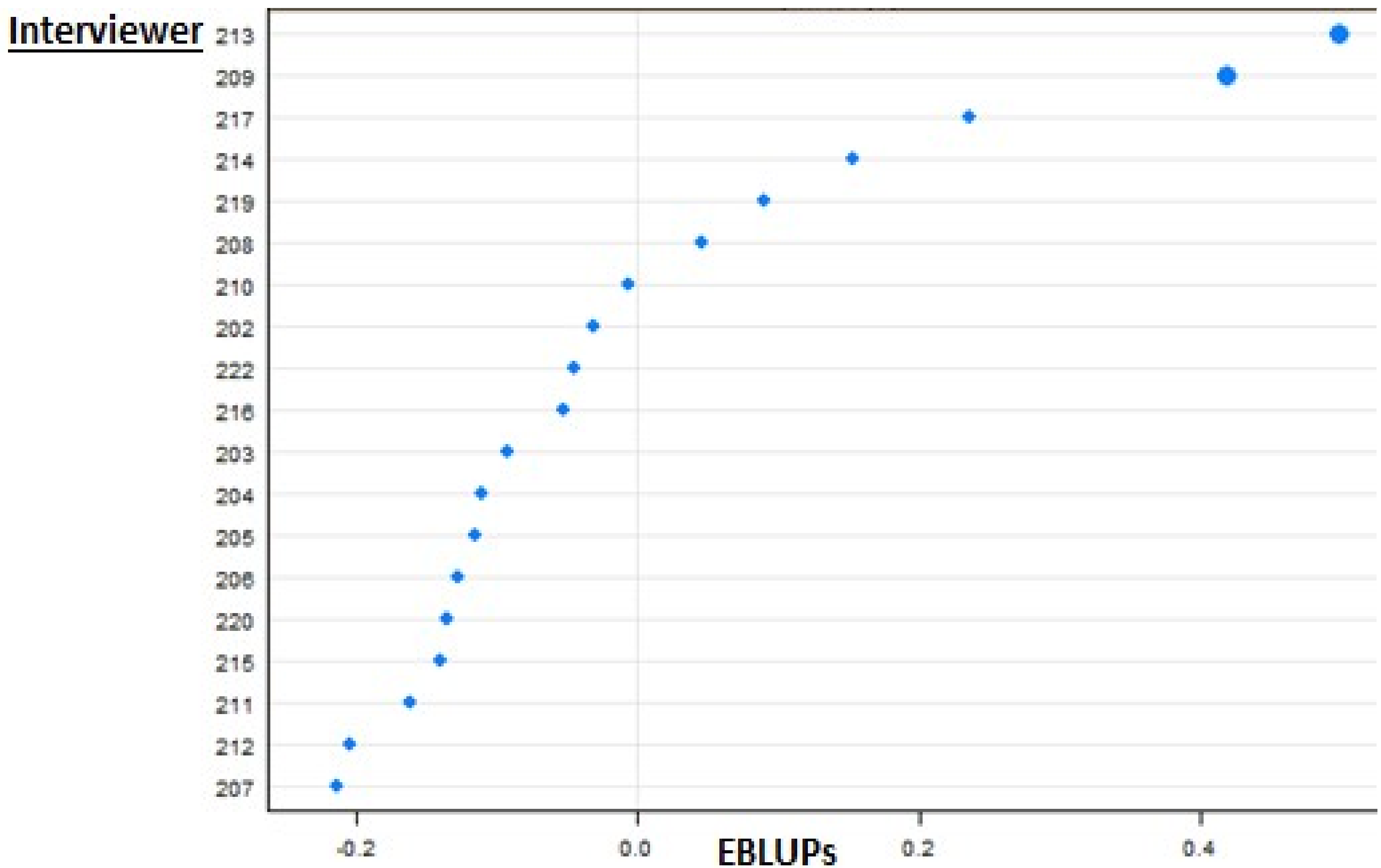
- Falsification

Sharma, S. & Elliott, M. R. (2020). Detecting falsification in a television audience measurement panel survey. *International Journal of Market Research*, *62*(4), 432-448.



These two interviewers' workloads were also geographically close to each other.

Interviewer-level Empirical Best Linear Unbiased Predictors (EBLUPs)



Many forms of interviewer errors leading to measurement error

- Falsification
- Incorrect inclusion/exclusion of persons in the Roster

Roster errors

- The household Roster is most often the first module in an instrument.

- Incorrect inclusion or exclusion of household members can have a major impact, e.g., household per capita income → policy could be wrongly influenced.

- Interviewers often given (or, should be given) extensive training on administering the Roster including definitional issues. Key: Who constitutes a household member ?

Example from the India Household Development Survey (2022-2024)

IHDS3_Roster | Prior_Wave_HH_Roster

Now, I want to make sure that I have the correct number of household members and their details such as name and age. Household members are those who have been living here under the same roof **and** shared the same kitchen for at least 6 of the last 12 months. Do not include non-resident members, relatives or temporary visitors who have lived here for less than 6 of the last 12 months.

Include servants, paying guests and other relatives who meet this definition. Also, include newly-married women or new-born babies.

Exclude married-out daughters.

Start with the head of the household and follow the specified sequence.

1. Enter [1] to continue

Roster issues…

- Beamon and Dillon (2012) experiment with 4 different definitions:
  1. Living in same space + same household head
  2. 1.+ Eat meals together
  3. 1. + common agriculture/income generation
  4. 2. + 3.
- Hypothesis: More restrictive definitions would produce smaller units.
- One among many findings : Definition 3 tends to increase household size vs Definition 1.
- Important that interviewers understand the correct definition and read/communicate it to the respondent accurately.

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Start with the head of the household and follow the specified sequence.

1. Enter [1] to continue

Helper tab comes up only after IWER completes the basic roster.

Many forms of interviewer errors leading to measurement error…

- Falsification
- Incorrect inclusion/exclusion of persons in the Roster
- Misunderstanding concepts or definitions
- Reading issues: Not reading the question fully, Misreading the question, etc.
- Probing issues: Not probing enough, Directed probing, etc.
- Not clarifying the respondent’s question adequately
- Inappropriate feedback
- Data entry errors ……………etc.

What is the impact of these differential behaviors?

Interviewer Variance

- A correlation among measured values collected by the same interviewers
  - "correlated response error"
- Hansen-Hurwitz-Bershad (1961) provide a model…

Hansen-Hurwitz-Bershad (1961) Model

$$y_{ijt} = X_i + M_{ij} + \varepsilon_{ijt}$$

where y_{ijt} is the response of the i-th reporter to the j-th interviewer on the t-th trial;

M_{ij} is the effect of the j-th interviewer on the i-th reporter; and

ε_{ijt} is the reporter- specific error on the t-th trial

- M_j is seen as varying over interviewers, but its expected value across all interviewers is zero

Kish (1962) Model

- Combines the true value (X_i) and the random error term (ε_{ijt}) into one term

$$y_{ijt} = X_{ijt} + M_j$$

where $X_{ijt} = X_i + \varepsilon_{ijt}$

Variance in Kish model

- Kish's intraclass coefficient

$$\rho_{\mathrm{int}} = \frac{Var(M_j)}{Var(M_j) + Var(X_{ijt})}$$

- Interpretation of Kish's ρ_{int} : ratio of variance between interviewers to the total variance of a measure

Estimating Interviewer Variance

- Kish' ρ_{int} is more widely used measure in practice
- Assuming interpenetrated design
- Historically: use ANOVA
  - More modern methods available

$$y_{ij} = \beta_0 + u_{0i} + \mathbf{X}_{\mathbf{ij}}^{\mathbf{T}} \boldsymbol{\beta}_{\mathbf{X}} + \epsilon_{ij}$$

$$u_{0i} \overset{iid}{\sim} N(0, \sigma^2_{iwer})$$

$$\epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2_e)$$

$$u_{0i} \perp \epsilon_{ij}$$

Also see:

The anchoring method: Estimation of interviewer effects in the absence of interpenetrated sample assignment. Elliott, West, Zhang, and Coffey, Survey Methodology, 2022.

Design Effect (deff)

- Inflation of the variance due to interviewers

$$Deff_{int} = 1 + \rho_{int} (m-1)$$

(m -> average workload)

- Effective sample size: n / deff

Variance Inflation Impact: $Deff_{int}$

	m=10	m=20	m=50
$\rho = .004$	1.04	1.08	1.20
$\rho = .009$	1.08	1.17	1.44
$\rho = .031$	1.28	1.59	2.52
$\rho = .072$	1.65	2.37	4.53

- Based on a model for the variance of an estimated mean
- High workload can exacerbate even small intra-interviewer correlations
- Reduction of interviewer workload can lead to lower interviewer effects. But assumption?

See for an LMIC example:
https://www.surveypractice.org/article/127468-can-i-interview-her-gatekeeping-in-a-telephone-survey-of-female-migrants-in-india

Practical Issues with Interviewer Variance Models

- Traditional models assume 100% RR and focus on measurement error.
- In practice, interviewers may successfully obtain cooperation from different pools of respondents (e.g., older vs. younger). Thus, nonresponse is another source of interviewer variance.
- Example: West and Olson (2010)
  - Variance for one variable (age at marriage) was largely driven by measurement error variance; variance for another variable (age at divorce) was largely driven by nonresponse error variance

Determinants of Rho

- Mode of interview
- Features of survey questions
- Interviewer demographics
- Interviewer behavior
- Interviewer training and supervision
- Respondent demographics
- …see Groves (Ch. 8) and West and Blom for a detailed discussion…

Mode of interview

- ρ_{int} is general lower in telephone than in face-to-face interviewers
  - Telephone (Tucker: .004; Groves & Magilavy: .009)
  - Face-to-face (Groves: .031)
- Possible explanation: less departure from script in centralized phone survey
- ρ_{int} is general lower in computerized interviews
  - Tourangeau et al. (.016 CAPI vs. .072 PAPI)

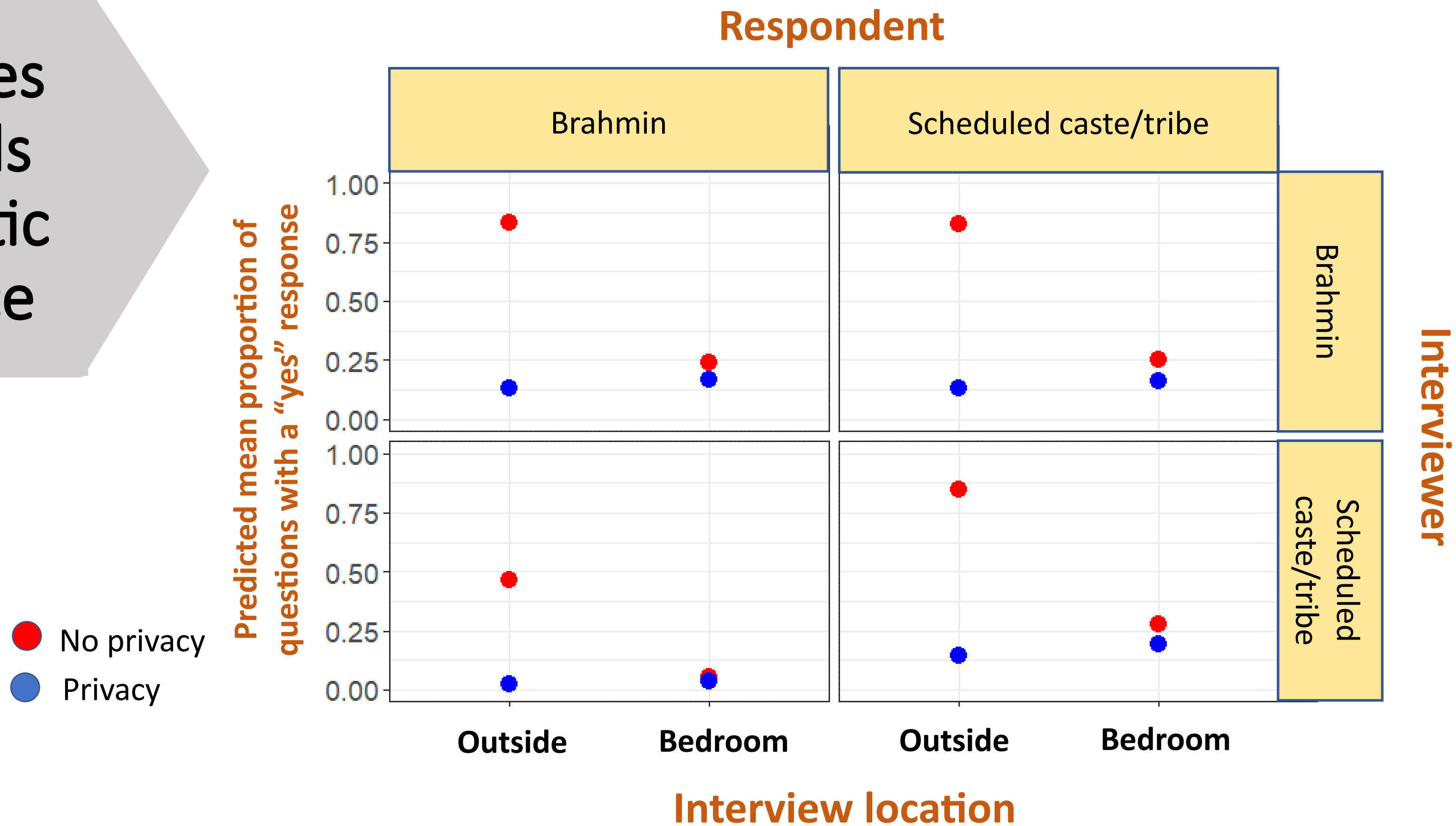
Features of survey questions

- ρ_{int} is generally highest for attitudinal, sensitive, ambiguous, complex, and open-ended questions (West and Blom)
- Explanation: These question introduce more opportunities for interviewers to intervene or assist respondents
- Groves 1989:
  - Open vs. closed: (.008 closed versus .012 open)
  - Factual vs. attitudinal: .0098 factual versus .0085 attitudinal

Interviewer demographics

- Interviewer demographics can produce interviewer effects when the measurements are related to these characteristics
- Example:
  - Race and racial items (Schuman and Converse, 1971)
  - Gender and gender-related items (Kane and Macauley, 1993)
- Explanations: social norms (politeness); priming

Attitudes towards Domestic violence



Sharma et al, 2023: PAA annual conference

The Effect of the Perception of an Interviewer's Race on Survey Responses in a Sample of Asian Americans (Liu, 2016)

- When respondents perceived the interviewers as Asian-American, they were more likely to show a preference for an Asian-American candidate in an election and to respond that Asians shared political interests.

- When respondents perceived the interviewers as non-Asian, they were more likely to admit that they had experienced discrimination.

- When Rs perceived IWERs as African-American, they were more likely to report that Asian Americans had things in common with African Americans.

Race-of-Virtual-Interviewer Effects (Conrad et al., 2019)

- Web survey using animated virtual interviewers (VIs)
- VI race mattered but not gender
  - When VI = ‘white”, 65.2% say “strongly oppose preference in hiring and promotion for blacks” vs 58.9% when VI = “black”

Interviewer behavior

- Few empirical evaluations between interviewer behavior and ρ_{int}
- Probing behavior appears to be a source of interviewer variance
  - Groves: probing responsible for variation in number of answers provided to open *Q*
  - Fowler and Mangione: failure to probe open *Q* was largest correlate of ρ_{int}
- Interviewer experience and expectations may drive behavior

Standardized Interviewing?

- General guidelines:
  - interviewers always read question exactly as it is worded; only administer neutral probes, only provide scripted feedback
- Prevailing philosophy and practice of collecting survey data
  - goal is for every respondent to get the same stimulus, and thus to remove the interviewer as source of error; should promote comparable data
  - Be aware of cross-cultural sensitivities

Evaluation of Standardized Interviewing

- In practice, interviewers deviate from guidelines
  - Up to 14% of questions read with major changes (Brenner, 1982)
- May not sufficiently motivate or inform respondents
- Guidelines may be incomplete
  - do not cover ALL possible interview situations
  - to remedy, would need to identify ALL of the interview situations that affect measurement error (implausible)

Interviewer training and supervision

- Fowler and Mangione (1985): Additional training (e.g., 10-day versus 2-day training) yields greater compliance with interviewer guidelines, but has no significant effect on ρ_{int}
- Tighter supervision associated with lower ρ_{int} (recordings may be source of improvement)

Summary of Determinants

- Many possible determinants have been tested
- Evidence suggests multiple causal mechanisms
- Lack of research on relationship between interviewer guidelines, training, and behavior and ρ_{int}

Respondents

Respondents as source of meas. error

Four basic factors (Sudman & Bradburn):

- Memory
- Motivation
- Communication
- Knowledge

1. Memory – Encoding and Forgetting

- Material may not be encoded in long-term memory to begin with
  - Lee et al., 1999: low levels of accuracy in parental reports on vaccinations - reflect poor initial encoding rather than retrieval failure.

- Material may be forgotten. In general:
  - The more time that has passed since the event occurred, the weaker the memory
  - The less distinct the event, the weaker the memory

Memory…

- Different retrieval strategies may lead to different errors. Three broad strategies:
  - Recall and count
  - Rate-based estimation (generally, overestimation)
  - Impression-based estimation
- The time at which something happened may be remembered incorrectly (i.e., telescoping)
  - Forward : events from outside the reference period are brought forward into the reference period
  - Backward : events from inside the reference period are taken out of the reference period

2. Motivation – Satisficing

- *R*s may be not be motivated to answer questions carefully and thoughtfully, especially when the task is difficult for them
- Potential consequences:
  - Response order effects
  - Acquiescence
  - Non-differentiation (or straightlining)
  - ‘Don’t know’ responses

Krosnick (1991)

- Described three determinants of satisficing:
  - Task difficulty
  - Respondent ability
  - Respondent motivation

$$p\,(\text{Satisficing}) = \frac{a_1(\text{Task difficulty})}{a_2(\text{Ability}) \times a_3(\text{Motivation})}$$

Task difficulty is within the researcher's *control*

3. Communication– Socially Desirable Responding

- *R*s may not tell the truth because they want to present themselves in a favorable light of fear of consequences
- Potential consequences:
  - underreporting of sensitive information
  - overreporting of socially desirable information

And even before that…

- May not understand what is being asked because it has technical, unfamiliar terms
  - Example: *Do you have any investments in an IRA?*

- May understand it in a different way than intended by researchers because of lexical ambiguity
  - Examples: *bank, employed, very often, pretty often*

Communication…

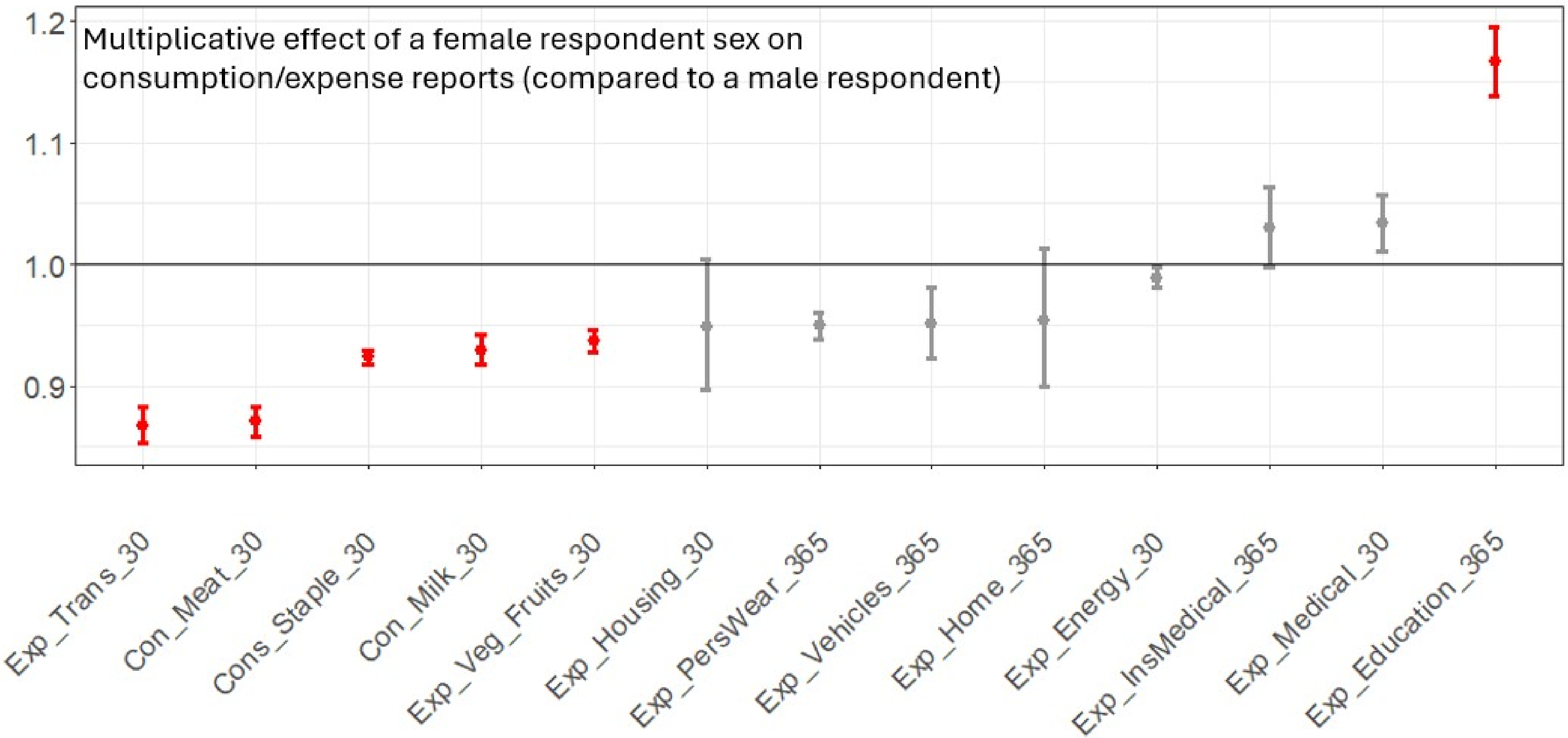
- Conversational interviewing: IWER and R work together to assure R understands the factual question as intended (Conrad and Schober, 2000)
  - Interviewer reads question as worded
  - Interviewer says whatever is necessary to assure R interprets question as intended, i.e., to ground meaning
- Goal is to standardize meaning, even if wording varies
- Tends to improve respondent comprehension but take longer in practice

4. Knowledge

- They may not know the answer to the question, and answer it without indicating their lack of knowledge
  - e.g., respondents report having attitudes about fictitious issues (Converse 1970)
  - "Nonattitudes": attitude is not crystalized and accessible

- Attempts to correct for its presence:
  - Explicit Don't Know (DK) options (*this can backfire*)
  - Strength of attitudes questions

Household consumption and expense questions: Male versus Female respondents in India



From Sharma et al, forthcoming

Questionnaire

Questionnaire – General wording Issues

- Instructions
  - Conflicting or inaccurate instructions
  - Complicated introductions or instructions
- Clarity
  - Question wording is lengthy, awkward, ungrammatical, or contains complicated syntax
  - Vague or technical terms
  - Reference periods are missing or not well specified
- Assumptions
  - Assumes constant behavior for situations that vary
  - Incorrect assumption made about *R* or about their living situation
  - Double-barreled

Willis (2005)

Instrument – Usability issues

- Questionnaire my have usability issues related to its layout or design
  - Difficult to read (e.g., small font)
  - Difficult to record response (e.g., small radio buttons)
  - Difficult to navigate (e.g., long scrolling page, auto-advance)
  - Programming errors (e.g., incorrect edit checks)
  - …


IHDS3 - Roster -- 10/20/2022 9:54 Am

IHDS INDIA HUMAN DEVELOPMENT SURVEY

IHDS3 - Roster -- 10/20/2022 9:54 Am

Suspend | DK | RF | Go To Roster | Previous Page | Next Page | Go To End | Go To Field | Hindi

नीचे दी गई सूची से SUKH RAM का चयन करें।

• सदस्य को विभिन्न नामों से जाना जा सकता है। सुनिश्चित करें कि आपने सही सदस्य का चयन किया है। इसे बहुत ध्यान से करें।

• यदि SUKH RAM का नाम सूची मे नहीं है , तो कोड "77" चुनें।

SUKH RAM के बारे में 2011 में ली गयी जानकारी से ठीक से मिलान करें:

○ 1. SUKH RAM 2011 में शादीशुदा पुरुष , 47 वर्ष की उम्र और मुखिया के रूप में रिपोर्ट किया गया है।

○ 2. PREMA DEVI 2011 में शादीशुदा महिला , 42 वर्ष की उम्र और SUKH RAM पत्नी/पति के रुप में रिपोर्ट किया गया है।

◉ 77. नया सदस्य

Roster ID	डिलीट	नाम	परिवार के मुखिया से संबंध	लिंग	जन्म का वर्ष	उम्र	वैवाहिक स्थिति	पूर्व सर्वे में नाम	अन्य सदस्य जोड़ें
1		SUKH RAM	1	1	1980	42	1	77	

Sample ID: A010013001601R | VD: 20-10-2022 | VT: 9:54:00 | CD: 8-2-2022 | CT: 11:6:30 | V.1.6 | ID.RosterBlock[1].RO1PriorID | Lat: 28.6242076999586 | Long: 77.2466679949357 | CTT | WW


IHDS3 - Roster -- 10/20/2022 9:54 Am

IHDS INDIA HUMAN DEVELOPMENT SURVEY

IHDS3 - Roster -- 10/20/2022 9:54 Am

Suspend | DK | RF | Go To Roster | Previous Page | Next Page | Go To End | Go To Field | Hindi

नीचे दी गई सूची से ALISOR का चयन करें।

• सदस्य को विभिन्न नामों से जाना जा सकता है। सुनिश्चित करें कि आपने सही सदस्य का चयन किया है। इसे बहुत ध्यान से करें।

• यदि ALISOR का नाम सूची मे नहीं है , तो कोड "77" चुनें।

ALISOR के बारे में 2011 में ली गयी जानकारी से ठीक से मिलान करें:

○ 17. SANIYA 2011 में अविवाहित महिला , 5 वर्ष की उम्र और ALISOR भतीजा-भतीजी के रुप में रिपोर्ट किया गया है।

○ 18. MAHESWAR 2011 में अविवाहित महिला , 3 वर्ष की उम्र और ALISOR भतीजा-भतीजी के रुप में रिपोर्ट किया गया है।

○ 19. ISRAR 2011 में शादीशुदा पुरुष , 28 वर्ष की उम्र और ALISOR भाई/बहन के रुप में रिपोर्ट किया गया है।

Roster ID	डिलीट	नाम	परिवार के मुखिया से संबंध	लिंग	जन्म का वर्ष	उम्र	वैवाहिक स्थिति	पूर्व सर्वे में नाम	अन्य सदस्य जोड़ें
1		ALISOR	1	1	1980	42	1		

Sample ID: A010013001501R | VD: 20-10-2022 | VT: 9:54:00 | CD: 8-2-2022 | CT: 11:6:30 | V.1.6 | ID.RosterBlock[1].RO1PriorID | Lat: 0 | Long: 0 | CTT | WW

Questionnaire – Framing

Framing of question may affect responses

- *global warming* vs. *climate change*; 44% of republicans endorsed former as real whereas 60% endorsed latter as real (Schuldt et al. 2011)

- *forbid* vs. *allow* public speeches in favor of communism, 39% yes to former; 56% no to latter (Schuman and Presser 1981)

Questionnaire – Response options

- Response options may affect their responses
- Issues include:
  - Number of response options
  - Agree-disagree vs. construct-specific scale
  - Low frequency vs. high frequency categories

Example: TV viewing

From Schwarz, Hippler, Deutsch and Strack's (1985) study: "How much TV do you watch daily?"

- 16% of respondents reported watching 2.5 or more hours of TV per day in low-frequency categories (up to ½ hour; ½ to 1 hour; 1 to 1 ½; 1 ½ to 2 hours; 2 to 2 ½ hours; 2 ½ or more)
- 37% percent of respondents reported watching that much TV in high-frequency categories (up to 2 ½ hours; 2 ½ to 3 hours; and so forth)

Questionnaire – Question Order

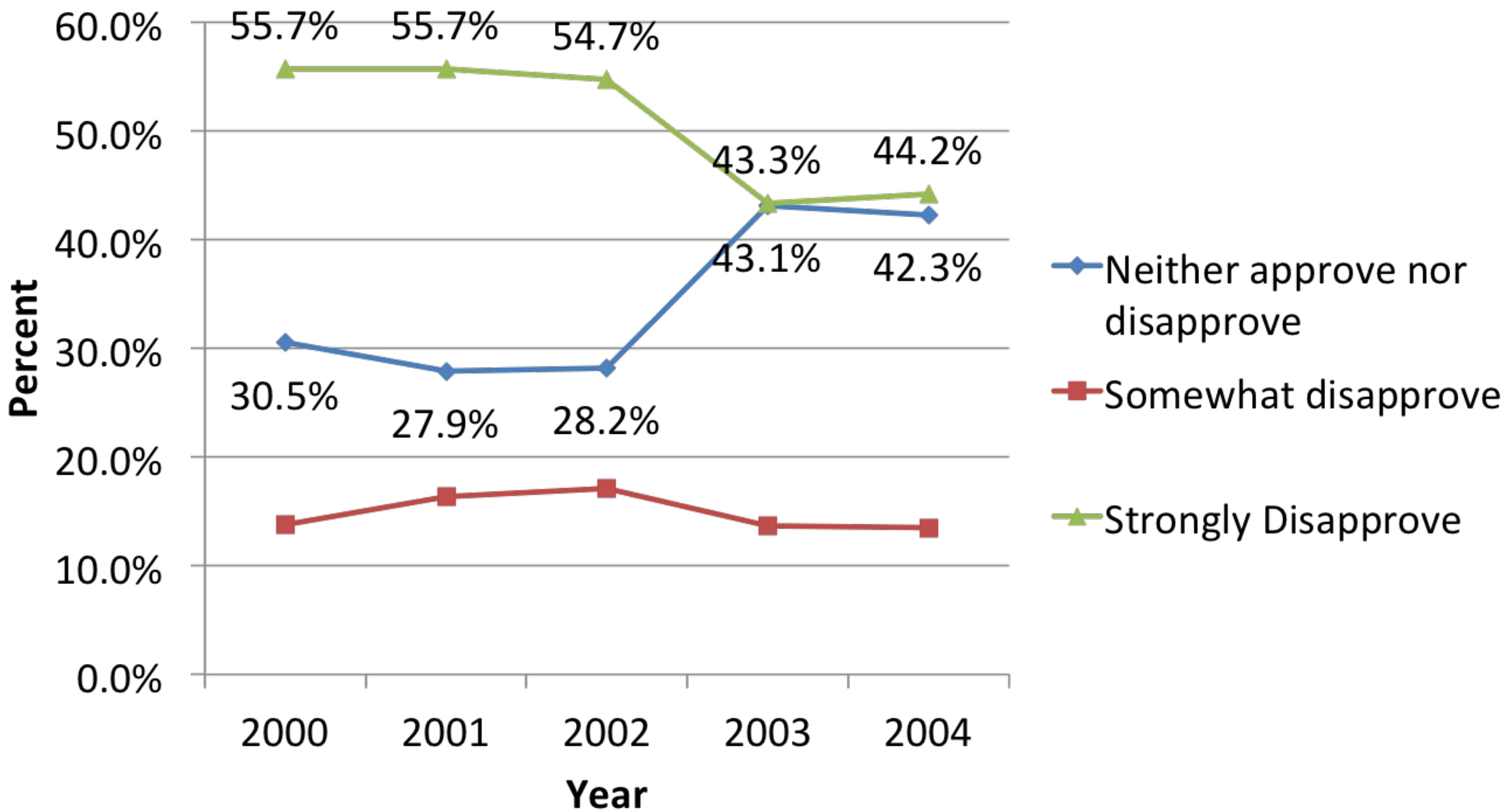
- Preceding questions may affect responses
  - Assimilation effect: accessible information from previous question is used to form representation of the construct being measured

    “happiness with your health” -> “happiness with life as a whole”
  - Contrast effect: accessible information used to construct a standard of comparison

    National Survey of Drug Use and Health:

    - through 2002, order was “how do you feel about cigarettes” -> “how do you feel about marijuana”
    - from 2003, order was “how many times have you attacked someone” -> “how do you feel about marijuana”

Attitudes Towards Trying Marijuana



Wang, K., R. Baxter, and D. Painter. (2005). Modeling Context Effects in the National Survey of Drug Use and Health (NSDUH). Proceedings of the Joint Statistical Meetings.

Questionnaire Design Recommendations - Krosnick and Presser (2010)

- Use 5- or 7-point scales
- Label all scale points
- Avoid agree/disagree, yes/no formats
- Do not provide DK option
- Move from more general to more specific

Recommendations Cont.

- When measuring amounts (e.g., “how many”; “how much”), use open questions
- Ask contingent items only after all the screening questions have been administered
- Reuse effective questions from earlier surveys
  - e.g., wwwn.cdc.gov/qbank

Next Week

- Processing issues in surveys
- Reading assignment
- Questions?